

# Chidiebere Nelson Ezi

*Data Analyst | MSc Data Science & Big Data Analytics*

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## PROFESSIONAL PROFILE

MSc Data Science & Big Data Analytics candidate at Wrexham University with a strong foundation in machine learning, statistical analysis, and data visualisation. Experienced in Python, SQL, and Tableau/Power BI, with hands-on project work spanning customer segmentation, deep learning image classification, dimensionality reduction, and AI-powered SQL agents. Eager to apply analytical skills in a graduate Data Analyst role to drive data-informed decision-making.

## TECHNICAL SKILLS

**Languages:** Python, SQL

**ML & Data Science:** Scikit-learn, Keras / TensorFlow, LangChain, PCA, K-Means, Random Forest, CNN

**Visualisation & BI:** Tableau, Power BI, Matplotlib, Seaborn

**Tools & Platforms:** Jupyter Notebook, PostgreSQL, LangSmith, Keras-Tuner, Git / GitHub

**Concepts:** Machine Learning, Deep Learning, Transfer Learning, NLP, Dimensionality Reduction, Data Wrangling, EDA

## PROJECTS

### AI SQL Agent on a Live Cloud-Based Database | *Data Science Infinity*

- Built an end-to-end AI SQL Agent using LangChain and GPT-4.1 on a live PostgreSQL database, enabling non-technical stakeholders to query data using plain English.
- Engineered a dedicated SQL system prompt and controlled schema exposure to ensure safe, accurate, and efficient query generation across multi-table joins and aggregations.
- Validated agent runs in LangSmith, confirming correct handling of grouping, aggregation, and date-range constraints — effectively creating a self-serve analytics layer over the database.

### Fruit Classification Using a Convolutional Neural Network | *Data Science Infinity*

- Designed and iteratively refined a CNN pipeline using Keras, progressing from a baseline 2-layer network to an optimised 3-layer architecture tuned with Keras-Tuner.
- Applied Dropout and Image Augmentation to eliminate overfitting, improving test-set classification accuracy from 75% (baseline) to 95% with the optimised architecture.
- Implemented Transfer Learning using the pre-trained VGG16 network, achieving 98% accuracy in just 10 epochs versus 50 epochs for custom models — demonstrating a clear business trade-off between performance and inference overhead.

### Customer Segmentation — "You Are What You Eat" | *Data Science Infinity*

- Developed a behavioural segmentation framework using K-Means clustering on grocery transaction data to support targeted marketing for a supermarket client.
- Applied the Within-Cluster Sum of Squares (WCSS / Elbow Method) to determine the optimal number of clusters, identifying 3 distinct, actionable customer segments.
- Surfaced key commercial insights: a mainstream 'Omnivore' segment (73.6%), a 'Hypothesised Vegetarian' segment (14.6%), and a 'Hypothesised Vegan' segment (11.8%) — directly informing customer communications strategy.

### Compressing Feature Space for Classification Using PCA | *Data Science Infinity*

- Applied Principal Component Analysis (PCA) to a high-dimensional dataset (100 features, 356 customers) to reduce the feature space while retaining 75% of explained variance, cutting features from 100 down to 24.
- Trained a Random Forest Classifier on the compressed feature set to predict album purchase propensity, achieving 93% classification accuracy.
- Delivered a proof-of-concept predictive model ready for scaling with live listening data to optimise targeted marketing spend.

## EDUCATION

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### MSc Data Science & Big Data Analytics

2024 – 2026 (Expected)

Wrexham University

**Relevant Modules:** Applied Data Science · Advanced Data Analysis & Visualisation · Database Systems & Data Analytics · Advanced Machine Learning

### BSc Mathematics

Undergraduate study — further details available on request

## ADDITIONAL INFORMATION

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**Certifications:** Data Science Infinity (Applied Machine Learning, Deep Learning, NLP, SQL Agent Development)

**Interests:** AI & LLM applications, sports analytics, continuous learning in data engineering